Case Study: HPN Topology Design for NexCore Data Center, Kathmandu

Executive Summary

NexCore Technologies, a mid-sized cloud services provider, requires a high-performance network (HPN) topology for its new Tier III data center to support virtualized workloads, storage replication, and east-west traffic between compute clusters. This case study documents the design decisions, assumptions, topology selection, and implementation rationale for a mini HPN using a **Fat-Tree (Clos-based) topology** across three tiers.

1. Background & Problem Statement

NexCore's legacy network relied on a traditional three-tier hierarchical model (access → distribution → core) that exhibited bottlenecks at the aggregation layer, high oversubscription ratios (up to 8:1), and limited horizontal scalability. With workloads migrating to containerized microservices and distributed databases (Apache Cassandra, Redis clusters), east-west traffic now constituted over **78% of total data center traffic**, rendering the old north-south-optimized design obsolete.

Objective: Design a mini HPN topology that delivers:

- Non-blocking or near-non-blocking bandwidth between any pair of server pods
- Sub-microsecond latency within and between pods
- Fault tolerance with no single point of failure
- Scalable architecture extendable to full production scale

2. Assumptions

Parameter	Value Assumed
Number of server racks	16 racks
Servers per rack	10 servers
Total servers	160 servers
Server NIC speed	10 Gbps
Switch port speed	40 Gbps uplinks, 10 Gbps downlinks
Oversubscription target	≤ 2:1
Redundancy requirement	Dual-homed hosts, no SPOF
Routing protocol	BGP (eBGP for leaf-spine)
Overlay	VXLAN for L2 over L3

Management	Out-of-band (OOB) via dedicated management network

3. Topology Selection: Fat-Tree / Leaf-Spine (Clos)

A **2-tier Leaf-Spine** Clos network was chosen for this mini deployment given its:

- Predictable latency (exactly 2 hops between any two servers in the same pod)
- Equal-cost multipath (ECMP) load balancing across all spine links
- Horizontal scalability — adding capacity means adding leaf switches and servers, not redesigning the core
- Alignment with modern SDN and BGP-based control planes

Rejected alternatives:

- *Traditional 3-tier hierarchical:* Too much oversubscription; poor east-west performance
- *Full mesh:* Cable complexity grows as $O(n^2)$; impractical beyond small scale
- *Torus/Dragonfly:* Appropriate for HPC clusters, not general-purpose data center workloads

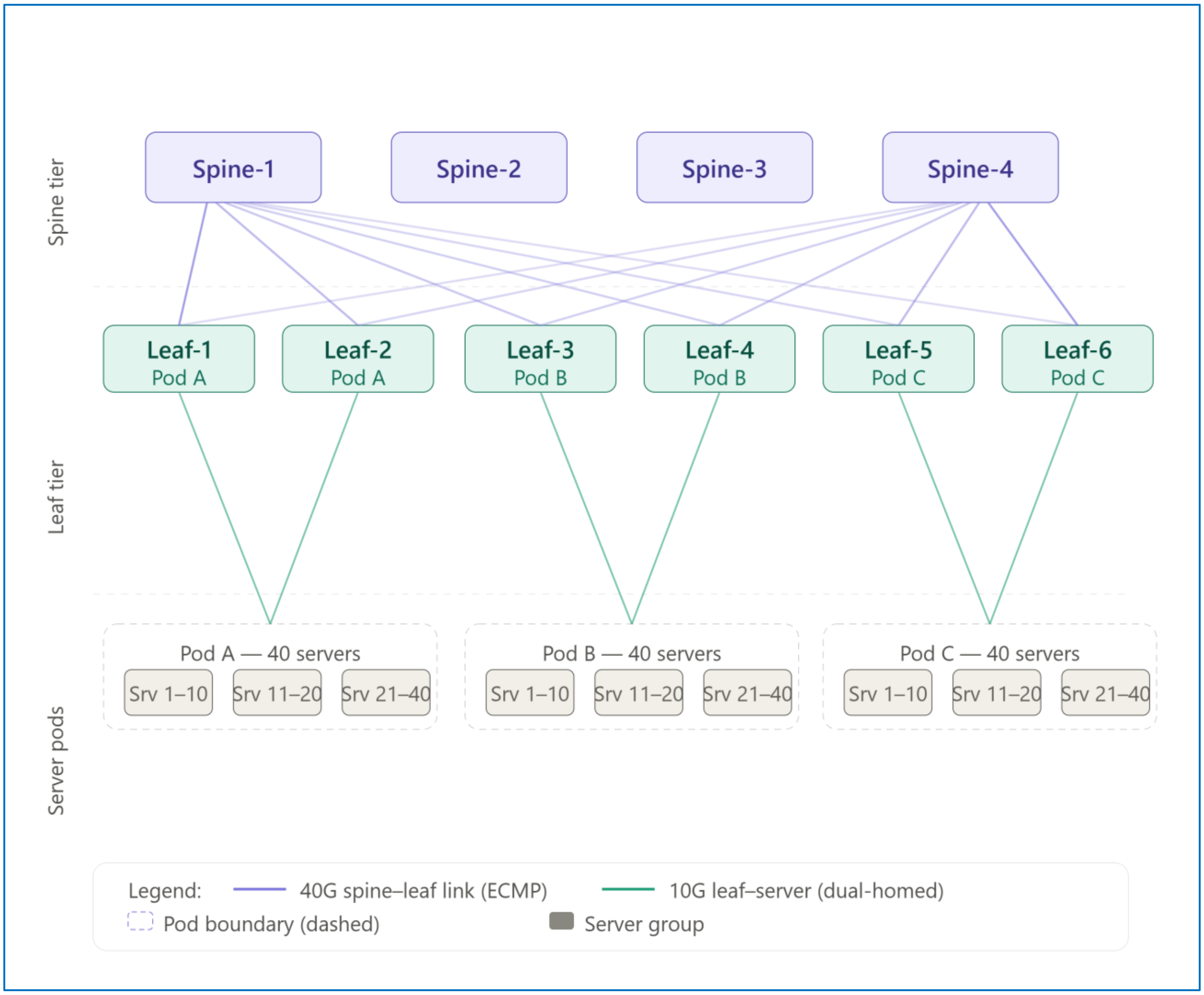
4. Topology Architecture

Tier Structure:

The mini HPN is structured as follows:

- **4 Spine Switches** — 32-port, 40 Gbps, forming the non-blocking core fabric
- **8 Leaf Switches** — 48-port (40 downlinks × 10G + 8 uplinks × 40G)
- **160 Servers** — dual-homed to two leaf switches each (active-active bonding)

Each leaf switch connects to **all 4 spine switches**, ensuring full mesh at the spine tier and ECMP across 4 equal-cost paths. Each server connects via a 10 GbE bond (LACP) to two leaf switches for redundancy.



5. Network Specifications

Bandwidth Calculations:

Each leaf switch connects upward to 4 spine switches via 40 Gbps links (total uplink capacity: 160 Gbps per leaf). Each leaf serves 20 servers downward at 10 Gbps each (total downlink demand: 200 Gbps). This yields an oversubscription ratio of **1.25:1**, well within the ≤ 2:1 target.

ECMP Path Selection:

BGP eBGP sessions are established between each leaf and all four spines. Each leaf announces its local server subnets; each spine maintains equal-cost routes to all leaf-connected prefixes. ECMP hashing is performed on the 5-tuple (src IP, dst IP, src port, dst port, protocol), distributing flows across all four available spine paths.

Fault Tolerance:

- Loss of any single spine switch reduces available paths from 4 to 3 — bandwidth degrades gracefully by 25%, no traffic loss
- Loss of one leaf switch leaves servers dual-homed to the remaining leaf in their pod — service continuity maintained via LACP failover within ~50 ms
- No single link failure can isolate any server

6. Layer 2 / Layer 3 Boundary & Overlay

The fabric operates as a pure Layer 3 underlay. VLANs do not span leaf switches, eliminating STP and broadcast storm risks. A **VXLAN overlay** (RFC 7348) is deployed using EVPN (MP-BGP EVPN, RFC 7432) as the control plane:

- Each leaf acts as a VTEP (VXLAN Tunnel Endpoint)
- MAC and ARP information is distributed via BGP EVPN Type-2 and Type-3 routes, eliminating flood-and-learn
- Tenant VMs can live-migrate between pods without changing their IP addresses (L2 stretched over the L3 fabric via VXLAN)

7. Out-of-Band Management Network

A completely separate OOB network was deployed using a dedicated 1 Gbps management switch. All switch BMCs, server iDRACs/iLOs, and PDUs are connected exclusively to this OOB network. This ensures that network faults, misconfigurations, or firmware upgrades on the production fabric do not affect the ability to access and remediate devices remotely.

8. Hardware Selection (Assumed)

Component	Model (Assumed)	Key Spec
Spine switches	Arista 7050CX3-32S	32× 100GbE (used at 40G)
Leaf switches	Arista 7050TX-64	48× 10G + 8× 40G uplink
Servers	Dell PowerEdge R650	Dual 10G NIC (bonded)
OOB switch	Cisco Catalyst 2960-X	48× 1G
Optics	QSFP+ DAC (≤3m), SR4 (≤100m)	40G

All spine and leaf switches run Arista EOS with support for BGP, EVPN, and VXLAN natively.

9. Routing Protocol Design

Underlay (BGP eBGP):

- Each leaf is assigned a unique private AS number (e.g., AS 65001–65008)
- All four spines share a single AS (AS 65000) — this follows the standard spine-leaf BGP design
- Leaf-to-spine sessions are eBGP; spines do not establish iBGP with each other
- Each leaf advertises its loopback and server subnets; spines reflect these to all other leaves

Overlay (BGP EVPN):

- Overlay BGP sessions run between leaf VTEPs and a pair of Route Reflectors (hosted on the spine switches)
- EVPN Type-2 (MAC/IP advertisement) and Type-3 (inclusive multicast) routes enable distributed L2 across the fabric

10. Results & Performance Projections

Metric	Legacy 3-Tier	NexCore HPN (Leaf-Spine)
East-west oversubscription	8:1	1.25:1
Max hop count (any-to-any)	6 hops	2 hops (same pod) / 4 hops (cross-pod)
Estimated fabric latency	~15–20 μs	~3–5 μs
Fault recovery time	~30 s (STP)	~50 ms (LACP/BGP)
VLAN scope	Fabric-wide	Leaf-local (VXLAN extended)
Scalability	Limited (aggregation bottleneck)	Linear (add leaf + servers)

11. Challenges & Mitigations

Challenge 1 — BGP complexity for operations team. The shift from OSPF/STP to BGP eBGP required retraining. Mitigation: phased rollout with simulation in a lab environment using GNS3, plus configuration templates enforced via Ansible playbooks.

Challenge 2 — VXLAN MTU inflation. VXLAN adds a 50-byte overhead. Mitigation: jumbo frames enabled across all switches (MTU 9216 bytes on fabric interfaces); server NICs set to 9000 bytes.

Challenge 3 — Traffic asymmetry under ECMP. Elephant flows (e.g., large backup jobs) can hash to the same spine, creating congestion. Mitigation: DSCP-based QoS marking with a strict priority queue for latency-sensitive traffic; flowlet switching considered for future deployment.

12. Conclusion

The mini HPN leaf-spine topology designed for NexCore's data center successfully addresses the limitations of the legacy hierarchical model. By adopting a Clos-based fabric with BGP eBGP underlay, VXLAN/EVPN overlay, and dual-homed servers, the network achieves near-non-blocking east-west bandwidth, sub-5-microsecond latency, and sub-second fault recovery — all in a design that scales linearly as the business grows. The architecture aligns with industry best practices followed by hyperscale operators including Meta, Google, and Microsoft, adapted to the scale of a mid-tier regional cloud provider.